Riders on the Storm*

Juan J Dolado[acd], Álvaro Jáñez[b] and Felix Wellschmied[ad]

[a]Universidad Carlos III de Madrid, [b]Stockholm School of Economics, [c]CEPR, [d]IZA

May 12, 2026

Abstract

Digital platforms often subcontract self-employed workers. We develop an equilibrium labor market model to quantify Spain's 2021 Riders' Law, which sanctions platforms for hiring self-employed couriers in the food delivery sector. Couriers in the model trade-off flexibility and employability as subcontracted against higher wages as employees. Counterfactual simulations show that, absent sanctions, one-fifth of employees would instead be self-employed. However, sanctions reduce welfare by lowering average hourly wages in the sector by 7 percentage points. We find that redistributing sanction revenue as lump-sum transfers to riders would offset most of these welfare losses.

Keywords: Riders, Food delivery platforms, Self-employed, Employees
JEL: J21, J60

*Corresponding author: Felix Wellschmied (fwellsch@eco.uc3m.es). We are very grateful to David Green, Belen Jerez, and participants in several seminars and workshops for useful comments, as well as to Ferrán Elías, Marta C. Lopes, Ana Moreno-Maldonado, Erik Öberg, and Clara Santamaria for helpful discussions at an early stage of this project. Financial support through grants RYC2023-043181-I, PID2024-158085NB-I00, and CEX2021-001181-M funded by MICIU/AEI/ 10.13039/501100011033 and by ERDF/UE is gratefully acknowledged.

1 Introduction

Like a dog without a bone, an actor out on loan, riders on the storm (**Riders on the Storm, The Doors**)

The digital platform economy is expanding rapidly worldwide, currently accounting for 4.4 to 12.5 percent of the global workforce (Datta et al., 2023). Because the degree of subordination of workers to platforms is difficult to determine, platforms often classify these workers as independent contractors rather than employees. Some see this behavior as a feature rather than as a bug, claiming that self-employment not only offers workers greater flexibility in terms of working hours but also facilitates their employability, as all it takes to start working is to sign up to a digital platform app. Yet, critics observe that the status of independent contractors takes away from workers essential social protections, such as being compensated for the number of hours actually worked and collectively bargained wages. Legislators worldwide have responded to these critics by proliferating new regulations governing the working conditions of platform workers.[1]

We contribute to this debate by analyzing the effects of a labor law reform in Spain that regulates the online food delivery sector. This sector has become the second-largest platform industry after taxi services, especially since the Covid-19 pandemic, which boosted the widespread use of these services globally. The so-called Riders' Law (RL, hereafter), approved in 2021, provides a particularly interesting setting by becoming one of the first enacted pieces of legislation that explicitly establishes the presumption of employment for food delivery couriers, commonly known as *riders*. A key feature of the Spanish setting is that some platforms resisted the new regulations in courts and continued to hire their riders as subcontractors.[2] As a result, these companies have been accumulating administrative fines that increased their relative operating expenses. Thus, as reflected by the lyrics of the classic song by The Doors heading this section, our riders, like theirs, seem to be weathering a storm full of uncertainties, this time related to their labor conditions.

To evaluate the consequences of the RL, we build an equilibrium model of the labor market that accounts for the above-mentioned key policy trade-offs. The economy features three types of jobs. First, there are regular (R) platform jobs, where riders are hired as employees. Since the hiring and vacancy-opening processes of these platforms

[1]For example, Australia (Fair Work Legislation Amendment Act 2024), Belgium (Wet van 28 november 2022), California (Assembly Bill 5 in 2019), Chile (Ley N° 21,431 in 2022), EU parliament (Directive 2024/2831), France (Loi 2016-1088), Italy (Legge 128/2019), Massachusetts (Ballot Question 3 in 2024), Mexico (Ley Federal del Trabajo 2024), Portugal (Lei 13/2023), South Korea (Employment Insurance Act 17,859 in 2021), or Uruguay (Ley N° 20,396 in 2025).

[2]Note that, since the RL only introduces a presumption, employers could provide evidence to the contrary by proving that their riders are not subordinated to the platform's guidelines. As discussed below, this explains why several platforms have challenged the new hiring regulations.

are similar to those in the non-platform economy, workers and employers are subject to standard search and matching frictions. These jobs offer fixed working hours, require employers to pay social security contributions, and pay hourly wages determined through Nash bargaining between a trade union and the employer. Second, there are frictionless casual (C) platform jobs performed by self-employed riders who freely choose their hours worked, pay their own payroll taxes, and are paid per delivered order. Compensation per delivered order, rather than per hour worked, is the rationale behind the immediate availability of these jobs. However, it comes at the cost of unpaid working time in terms of idle waiting between orders. Third, there is an exogenous outside option capturing riders' available job opportunities in the rest of the labor market. Given idiosyncratic preferences for hours worked, riders choose where to search for employment and how many hours to work when self-employed. Importantly, they always have the option of joining a C platform and search on-the-job for R platform jobs, if this is their preferred option. Finally, imperfect substitutability among consumers for riders' services across platforms ensures that both platforms operate in equilibrium and demand labor.[3]

We estimate the model parameters to fit post-reform moments regarding hours worked, wages, market shares, and firms' provisions for RL-related legal risks. To do so, we use microdata from a representative convenience survey, firms' annual reports, and aggregate statistics from the Spanish food delivery sector. Three key empirical facts guide the estimation approach: (i) riders with similar characteristics earn an hourly wage premium in R jobs compared to C jobs, (ii) a substantial fraction of C and R riders overlap in their hours worked, and (iii) some C riders work a relatively high number of hours. To reproduce these facts, the model implies that riders with low distaste for long working hours choose C jobs to earn higher earnings. Other riders search for R jobs due to their wage premium. Yet, due to search frictions, not all instantaneously find an R job and choose to work on a C platform in the meantime, leading to more than half of current C riders actually preferring to work for the R platform. These riders have potentially much to gain from a reform that expands employment at the R platform: due to the idle time waiting for orders in the C platform, almost all C riders work more hours than they would work at the R platform, but are paid for fewer hours and produce fewer orders.

The RL's aim is to achieve such an expansion of R jobs. It does so by increasing the operating costs among the platforms that keep hiring independent contractors through administrative sanctions. We estimate the size of the sanctions from non-compliant firms' provisions for RL-related legal risks between 2021 and 2024, which amount to 3.2 percent of their net revenue. These provisions reflect both imposed but not yet court-confirmed

[3]There is media evidence that consumers of food delivery services in Spain are not indifferent between different platforms and, e.g. show clear preferences between Glovo and Just Eat, based on restaurant availability, app functionality, price, and speed. Thus, while Glovo focused on fast, multi-product delivery, Just Eat is valued for its network of local restaurants and promotions (El Español, 2021).

administrative sanctions and prospective future sanctions. Importantly, instead of being fixed costs, these sanctions are proportional to the number of subcontracted self-employed riders due to size-dependent penalties and back-payment of avoided Social Security taxes. Therefore, sanctions increase the marginal cost of producing orders by C platforms, raising their relative prices and shifting consumers' orders demand to R platforms.

In that way, the reform accounts for nearly the full 6 percentage point shift in market shares of complying platforms we observe in the data. To satisfy this additional demand, R platform increase their employment share by 4.1 percentage points, which absorbs 80 percent of C job losses resulting from the law. We find that, in the absence of sanctions, one-fifth of employees would instead be self-employed. In partial equilibrium, the reallocation towards employees would be highly beneficial. As explained above, most C riders could work fewer hours at R jobs while receiving higher pay. To understand the general equilibrium effects, note that, as the C platform loses both riders and orders, the effect of the reform on idle waiting time is non-obvious. As those riders who switch to the R platform produce more orders at that platform than at their old C jobs, and the reform leads to a decline in the total number of orders, the remaining C riders must become less productive, i.e., their waiting time increases by 7 percentage points. The result is a similar decline in their hourly pay. Moreover, as conditions at the C platform worsen, riders' outside option in bargaining with the R platform also worsens, leading to a decline in hourly wages by 6 log points. All in all, we find that the food delivery sector contracts by about 0.3 percent and its average net hourly wage by 7 percent, leading to a reduction of both riders' and consumers' welfare.[4]

Finally, we study the effects of complementing the sanctions with transfers to riders. In the first experiment, the revenues generated by fines are returned to the riders through lump-sum rebates. Rebating sanction revenues to riders reduces their welfare loss from the RL from 7.2 percent in terms of consumption-equivalent variation to 2.5 percent. Moreover, welfare losses vanish when transfers reach 1.5 times sanction revenues, with the resulting deficit representing only a tiny portion of government public revenues. In contrast, when complementing the sanctions with a cut in the R employer's payroll tax, welfare is reduced by an additional 1.0 percent. The intuition is similar to the welfare loss from the RL: expanding R platform jobs increases waiting time at C jobs, which not only hurts workers who remain in those jobs but also deteriorates the bargaining position of R platform riders.

[4]Given that the actual number of riders has increased by 40 percent since the pandemic (see Sections 2 and 6 below), these negative effects of the RL should be interpreted as deviations from a growing trend in households' demand for online food delivery, which is taken as exogenously determined in our counterfactual simulations regarding steady-states comparisons.

1.1 Related literature and outline

Our paper contributes to a growing literature on the characteristics of the online gig economy. A large body of empirical research deals with the difficulty of measuring platform work arrangements through either conventional administrative data, specific surveys, or field experiments (Mas and Pallais, 2017; Katz and Krueger, 2019; Boeri et al., 2020; Bracha and Burke, 2021; Garin et al., 2022, 2023). To reduce measurement problems, our paper uses an own-elaborated convenience survey that aligns with representative aggregate information from the food delivery sector.

Similar to us, there is a recent literature that also highlights the high value that digital platforms provide to workers through flexible working arrangements. Chen et al. (2019) study the case of Uber drivers and show that their reservation wages vary substantially over time, i.e., they value flexibility in their working hours. As a result, their surplus is higher than in alternative work arrangements which offer less flexibility. Stanton and Thomas (2025) analyze a platform environment where buyers post one-time projects and workers compete for these projects by posting wages. They find that imposing traditional employment regulations on short-term platform work reduces overall welfare and workers' surplus. Key to this finding is that: (i) workers already capture a large share of the surplus, and (ii) the regulation decreases demand for their services. We add to this literature by studying an environment where flexible platform jobs coexist with workers being hired as employees, therefore highlighting the importance of platforms using independent contractors to overcome labor regulations. Moreover, we show that regulating firms with independent contractors has negative spillover effects on employees' wages when these are bargained. Nevertheless, our conclusions are similar: some workers value the flexibility that the independent contractor status provides, and attempts to regulate these types of jobs often lead to lower labor demand and an overall loss in workers' welfare.[5]

Scarfe (2019) and Dolado et al. (2025) study casual and zero-hour contracts in Australia and the UK, respectively, which are other work arrangements that are akin to those used for subcontracted riders. Under those work arrangements, firms only call up workers when they need their services, leading to workers being operative in some periods but not in others. Just like us, these papers employ structural search and matching models to analyze the general equilibrium effects of regulating flexible contracts. However, we differ from their approach in three dimensions. First, we introduce ex-ante worker heterogeneity. Second, we build a two-sector model. Third, the flexible C job features endogenous

[5] Angrist et al. (2021) study a yet different setup where drivers choose between permanent lease contracts (taxi driver) and revenue fees (Uber driver). In that case, the worker retains his work flexibility in both work arrangements.

wages through the determination of idle waiting time because platforms ensure that the demand and supply of riders match.

Our paper also relates to the literature that uses structural equilibrium models to study the informal economy, which resembles independent contractor arrangements in the platform economy. Part of this literature builds two-sector models that feature a frictional and a frictionless sector, such as Zenou (2008) and Satchi and Temple (2009), which model an economy with homogeneous workers who always prefer formal jobs. Instead, our approach is similar to Albrecht et al. (2009), Haanwinckel and Soares (2021), and Feng et al. (2024) by allowing for ex-ante worker heterogeneity to generate the possibility that some workers may prefer C jobs. Different from these papers, our sorting mechanism is based on the taste for hours worked rather than by ability. We show that the self-selection of workers into sectors by hours worked has key policy implications arising from the interaction of search frictions in the R sector and idle time in the C sector. Moreover, while these papers focus on the effects of taxes or transfers on formal-sector jobs, we study the effects of regulating C jobs and show that this type of regulation leads to spillover effects on wages and employment in R jobs through the importance of outside options for collective bargaining.

Finally, our paper also speaks to other strands of the literature that deal with the effects of changing legal work-time regulations (Carry, 2022), and the modeling of hours of work in search and matching models (Cooper et al., 2017; Frazier, 2018). We depart from these works in allowing for two-sided heterogeneity regarding labor demand and labor supply decisions made by firms and workers, respectively.

The rest of the paper is organized as follows. Section 2 provides institutional background about food delivery platforms in Spain, plus details of the 2021 Riders' Law. Section 3 describes the data sources and draws stylized facts about the riders' labor market outcomes. Section 4 lays out the quantitative model. Section 5 describes its calibration. Section 6 shows the counterfactual results from policy experiments. Section 7 concludes. An Appendix gathers further evidence discussed in the main text.

2 Institutional Background

As in many other developed countries, the Spanish online food delivery sector has expanded rapidly over the past decade. This expansion took place in a context of uncertainty about the correct classification of the employment status of riders which platforms have used to hire their riders as independent contractors rather than as dependent employees, with the former accounting for 81.2 percent of all riders by 2020. Hiring workers as independent contractors has several advantages for platforms. For example, they

do not need to comply with labor law on overtime schedules and avoid unionization of their workers. Moreover, employers do not pay social security contributions. Finally, by compensating workers for actual completed orders, rather than by a fixed hourly wage, employers shift all risk of insufficient demand to the worker.

The widespread practice of hiring workers as independent contractors has led to an intense debate about the appropriate regulation of work arrangements in the food delivery sector. As a result of this discussion, the Spanish parliament approved the RL (*Real Decreto-Ley 9/2021*) in September 2021. Regarding the motivations of policy-makers, the preamble of the RL states that the advantages from the information and communication technologies "are perfectly compatible with the objective of labor law to re-balance the interests of the different agents in this sector", and that these regulations should "guarantee that technological change provides positive effects equitably". To achieve these goals, the RL introduces: (i) the presumption that riders are employees rather than independent contractors, and (ii) the requirement that platforms disclose the rules underlying algorithmic decisions that affect workers' labor market conditions. Thus, in essence, the RL explicitly extends standard labor regulations for employees to food delivery riders, including more rigid work schedules and collectively bargained wages, supported by greater transparency in information.

Our analysis exploits a key feature of the RL, namely, the fact that the status of the rider is only a presumption that could be rebutted if the platform shows that it did not exert the powers of organization over the rider. Resulting from this remaining legal uncertainty, some platforms defied the new regulation on the obligation to hire riders as employees, while others complied with it. Consequently, the sector features riders with different employment statuses both before and after the reform.

In particular, three of the four dominant players (Glovo, Uber Eats, and Deliveroo), with a joint market share of two-thirds, continued hiring most of their riders as independent contractors, so we call them "defiers". In contrast, the remaining major platform (Just Eat), with one-third of the market and which had already hired riders as employees before the RL, decided to move a step forward by signing a collective agreement with the main trade unions in the sector, so we label it as "complier". Importantly, the legal arguments of the defiers did not convince the government, which responded with administrative sanctions for which the firms have provisioned approximately €923.5 million for the resulting legal risks. We interpret these provisions as the present discounted expected value of sanctions, which represents about 3.3 percent of defiers' net revenue (see Appendix B).

Indeed, market share data after the enactment of the reform is consistent with a deteriorating relative advantage of the defier platforms. Figure 1 shows that their market

Figure 1: Market shares in the Spanish food delivery sector

Source: We use information from Measurable AI (Spain Food Delivery Market Overview) for 2020 and 2021. For 2023, we use data from Dashmote (Spain's Food Delivery Market: An Analytical Overview)
Note: The graph displays the revenue market share in the Spanish food delivery sector between 2020 and 2023.

share declined by 6 percentage points within two years of the reform, despite having expanded previously. Partly, the decline results from the exit of Deliveroo from the Spanish market after conducting a mass-layoff of nearly $4,000$ riders. Though shifting employment to platforms hiring employees is most likely consistent with the goal of the law's drafters, some organized labor groups have opposed the reform. For example, according to one of the main workers' associations (*Asociación Autónoma de Riders, AAR*), about 65 percent of its members dislike the new regulations, mainly due to the rigidity of hours of work in the R platforms.

3 Data and Stylized Facts

This section summarizes the data sources and the main stylized facts of the Spanish food delivery sector that allow us to develop a quantitative model evaluating the effects of the RL. Given the scarcity of statistical information on this sector, we resort to an own-elaborated online convenience survey to obtain microdata on riders' labor market outcomes. Interestingly, a comparison of this dataset with aggregate data from alternative sources shows that our survey respondents closely resemble the broader riders' population in terms of key sociodemographic attributes. Next, we document three main findings: (i) higher net hourly wages are earned by R riders, controlling for sociodemographics, (ii) relatively long hours worked are only a feature of C riders, and (iii) short hours are common for both types of riders.

Table 1: Descriptive statistics

	Convenience survey (2023)	Adigital (2020)
Workers' characteristics	**Mean**	**Mean**
Age	27.3	29.3
Gender (Male)	0.86	0.89
Education (Upper)	0.46	0.37
Nationality (Foreign)	0.77	0.72
Work Permit (Yes)	0.82	0.75
No. of platforms (2023)	1.3	
Tenure (years)	1.5	
Wages and hours		
Net hourly wage (Euros)	5.6	
Daily hours	4.6	
Platform employment shares		
Defiers	0.76	
Compliers	0.24	

Note: Convenience survey data contains information about 162 riders via Google.form corresponding to the period September-October 2023. The Adigital (2020) contains information about 1,852 riders during the period January-February 2020.

Data. Through some personal contacts among riders from complier and defier platforms, we distributed a small convenience survey to riders in a completely anonymous format via the Google- Forms platform between Sep. and Oct. of 2023. The questionnaire was organized around three blocks: (i) general information about the worker (age, gender, educational attainment, nationality, and availability of a work permit, number of platforms, and tenure); (ii) information about their net hourly wages and daily hours worked; and (iii) platform shares (defiers and compliers). In total, 320 riders received the questionnaire, out of which 162 responded. Since approximately 30,000 riders were active in 2023 according to the AAR, our sample represents about 0.5 percent of the rider population. In addition, we complement this survey with aggregate data from a larger survey conducted by the Spanish Association of Digital Economics (Adigital, 2020), which collects information on similar sociodemographic characteristics for a representative sample of 1,852 riders in early 2020. Yet, unlike ours, this larger survey lacks information on wages and hours worked, as well as on the type of platforms riders work for.

Table 1 summarizes the main descriptive statistics of the Adigital (2020) survey (pre-reform) and the convenience survey (post-reform). Respondents in the convenience survey closely resemble the broader population of riders captured in the larger survey across key sociodemographic dimensions, including age, gender, education, nationality, and work permits.[6] This similarity supports the representativeness of our convenience survey, as

[6]Unfortunately, the microdata of the Adigital survey is not available. Thus, we cannot provide standard statistical tests about differences in means.

there are few reasons to expect that the RL has altered the composition of riders, particularly given the strong segmentation of this labor market. Riders are predominantly young, male, foreign, and their tenure is 18 months. Moreover, the average rider typically works for a single platform despite the possibility of sharing several platforms. Interestingly, about 18-25 percent of riders are migrants without a work permit who work for C platforms since it would be illegal for R platforms to hire them as employees. This may partly explain the resistance of the C platforms to comply with the RL.

Stylized facts about hours and wages. As described above, an advantage of being hired as an employee is being paid by the hour with a wage that is set by collective bargaining. Moreover, R platforms pay payroll taxes. To understand the effect of the employee status on net wages, our survey explicitly asks respondents to report their net hourly wage after deducting maintenance costs and payroll taxes. Using this information, we run the following *mincerian* OLS regression:

$$\ln w_i = \beta_0 + \beta \mathbb{I}_{C=1} + \gamma \boldsymbol{X_i} + \varepsilon_i$$

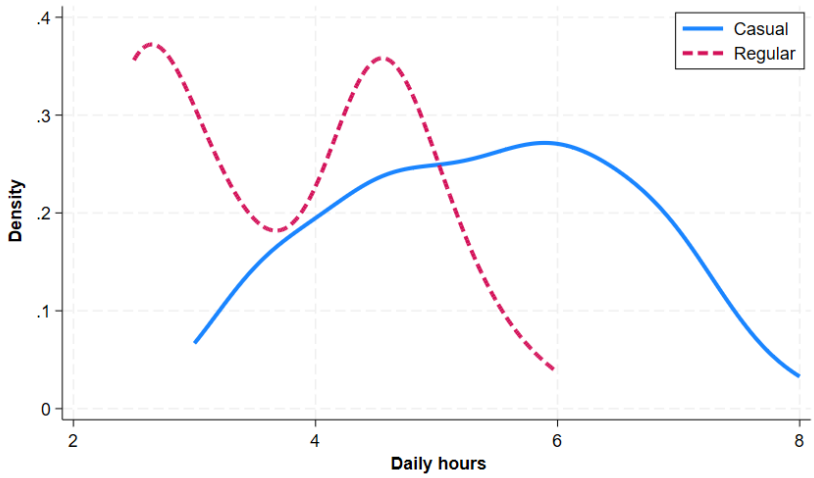
where $\ln w_i$ is (logged) net hourly wage, $\mathbb{I}_{R=1}$ is a dummy variable capturing if a rider works in platforms offering C jobs (Glovo or Uber Eats), and $\boldsymbol{X_i}$ is a vector of sociodemographic covariates.[7] We find that, on average, the hourly wage among C riders is about 17.2 log points lower than that of R riders with similar observable characteristics ($\beta = -0.172^{***}(0.033)$), a result which is consistent with trade unions being effective in extracting rents.

Note that a sizable share of riders (76 percent) opt for C jobs despite their lower wages. To understand this fact, Figure 2 displays the densities of daily hours worked in both types of jobs, where four findings stand out. First, there is a large dispersion in daily hours across riders, suggesting large heterogeneity in their time availability, which is consistent with reports from the Adigital (2020) survey. Second, the distribution of hours worked in C jobs is more dispersed than in R jobs; in particular, the density of hours worked of the former jobs is smooth, while it is bimodal for the latter.[8] Third, C jobs mainly provide upward flexibility in hours worked: while R riders work less than 6 hours per day, a significant proportion of C riders exceed this threshold, implying that the average daily hours worked in C is higher than in R (5.4 h. vs 3.7 h.), a result that also holds after controlling for observables. Fourth, the distribution of hours worked

[7] We include age, gender, nationality, work permit, tenure, and education.

[8] This result is consistent with Just Eat offering two main job contracts: one of 12 weekly hours during weekends and another one ranging from 16 to 30 hours during the entire week. However, for tractability, we assume in the model that R platforms offer a single work schedule (e.g. a weighted average of the two contracts)

Figure 2: Distribution of hours worked



Note: The Figure displays the density of hours worked for workers with a casual and regular platform job using a Gaussian Kernel estimate.

has a significant mass at both types of platforms at low hours worked, typically below 4 hours. We interpret this fact as consistent with the existence of search frictions in the R platform that ration the access of riders with strong preferences for these jobs due to their wage premium.

4 Quantitative Model

Environment. We evaluate the effects of the 2021 RL employing a quantitative steady-state equilibrium model in discrete time. The model incorporates the main trade-offs described above between letting heterogeneous platforms hire workers either as self-employed or as employees. We distinguish between: (i) platforms offering frictionless casual (C) jobs, which can be instantly found by self-employed workers who choose their optimal working hours freely; (ii) platforms offering frictional regular (R) jobs, where workers can only be hired as employees and are paid a collectively-bargained hourly wage; and (iii) an outside option, which represents job opportunities amenable for riders in the rest of the labor market. Furthermore, riders are ex-ante heterogeneous in their tastes for hours worked, determined by parameter ϵ (to be defined below). Finally, platforms sell their output to a representative consumer who views their respective services as imperfect substitutes. Thus, the model features both types of platforms operating in equilibrium with their market shares being endogenously determined.

Production technology. We consider two representative groups of platforms with identical production technologies, which are characterized by the type of job they offer,

$j = C, R$. The production of orders, o_j, is assumed to be linear in labor:

$$o_j = \int_0^{n_j} A h_j^e dj, \tag{1}$$

where n_j denotes employment, h_j^e are the effective hours worked (defined below), and A captures productivity per hour.

Labor market for regular jobs. The labor market for R jobs operates as in a standard Diamond-Mortensen-Pissarides setup, in which search and matching frictions hinder the formation of new matches. Riders are only hired as dependent employees, which implies that they earn collectively-bargained wages *per hour* worked, while employers pay payroll taxes τ_f. Hours worked at this platform are assumed to be fixed $\bar{h}_R$.[9] Searchers and R platforms meet according to a matching technology that determines the contact probabilities of riders and vacant jobs. To ensure that these probabilities are bounded between 0 and 1, we adopt the CRS matching function proposed by Den Haan et al. (2000), namely, $m(s, v) = sv/(s^\iota + v^\iota)^{1/\iota}$, where ι is a matching parameter. Hence, the contact probabilities for job seekers and for vacancies are $p(\theta) = m(s, v)/s$ and $q(\theta) = m(s, v)/v = p(\theta)/\theta$, respectively, where $\theta = v/s$ is labor market tightness, given by the ratio of vacancies, v, to searchers, s, where the latter encompasses both the unemployed and on-the-job searchers.

Labor market for casual jobs. The labor market for C jobs is fundamentally different from that of being an employee. That is, instead of having to wait for firms to post vacancies, workers have the option to take up employment instantaneously by registering as self-employed in the app. Moreover, instead of being paid by the hour, riders with a C job get paid *per order* produced. Note that paying riders by the order, instead of by the hour, implies that C platforms do not experience an increase in their costs when the riders' labor supply exceeds the market demand for their orders. The app, in turn, automatically aligns any labor supply with their demand for orders. As a result, workers suffer from lower hourly pay when the supply of total riders' hours exceeds the demand for orders, as part of their working time remains idle while waiting for an order to arrive. This experience of idle, unpaid waiting time is a key new phenomenon brought in by the rise in platform work throughout the world (see ILO (2024)).

For riders, the advantage of this work arrangement is that they can freely choose their hours worked in a decentralized fashion, which determines the total labor supply in the C platform. When there is excess supply of hours, the effective productive hours that a C

[9] As shown below, riders preferring short hours always choose work in R jobs due to their wage premium (see Figure 3). Thus, allowing for a part-time work schedule on the R platform does not affect worker sorting across jobs and, therefore, would not alter the main results of the paper.

rider gets paid is only a fraction of the supply of total hours: $h_C^e = \varphi h_C$, where $0 < \varphi \leq 1$ is an endogenous proportionality factor strictly lower than one when demand for hours falls short of supply. Hence, the market-clearing condition for the C platform is

$$o_C = \int_0^{n_C} A h_j^e dj = n_C \varphi A \int h_C(\epsilon)\, dG^C(\epsilon), \tag{2}$$

where G^C is the cumulative distribution function of the riders' taste parameter ϵ for the hours worked in these jobs, so that $\int h_C(\epsilon)\ dG^C(\epsilon)$ are the average hours worked per rider.

Demand for food delivery orders. We model the demand for food delivery orders through a representative consumer with exogenous income m who chooses the number of orders o_j from each platform by maximizing the following CES utility function subject to a linear budget constraint:

$$\mathcal{U}_c = \max_{o_C, o_R} \left(\sum_{j \in \{C,R\}} (s_j o_j)^{\frac{\mu-1}{\mu}} \right)^{\frac{\mu}{\mu-1}}, \tag{3}$$

$$\text{s.t.} \quad p_C o_C + p_R o_R = m. \tag{4}$$

where $\mu > 1$ is the elasticity of substitution between orders provided by C and R platforms, and s_j are utility shifters capturing the importance of non-price characteristics of food delivery services in platform j. Note that, for tractability, we abstract from other potential substitute goods in the utility function (e.g., cooking at home). The first-order conditions (FOC) of the consumer problem yield the relative inverse demand of orders:

$$\frac{p_C}{p_R} = \left(\frac{s_C}{s_R}\right)^{\frac{\mu-1}{\mu}} \left(\frac{o_R}{o_C}\right)^{\frac{1}{\mu}}, \tag{5}$$

where we normalize $s_C = 1$ since consumption choices only depend on the relative utility shifter.

Firm's problem. Firms choose the number of orders to maximize flow profits, having internalized their downward-sloping demand functions and taking as given both the riders' labor supply decisions and their competitors' decisions:

$$\max_{o_j} \ \Big(p_j(o_j) - mc_j\Big) o_j,$$

$$\text{s.t.} \quad p_j(o_j) = \left(\frac{s_j}{s_{-j}}\right)^{\frac{\mu-1}{\mu}} \left(\frac{o_{-j}}{o_j}\right)^{\frac{1}{\mu}} p_{-j},$$

with mc_j denoting the marginal cost of each platform j. The FOC yields the standard Lerner condition for the price as a markup over the marginal cost:

$$p_C^* = \frac{\mu}{\mu - 1} \cdot mc_C, \quad \text{and} \quad p_R^* = \frac{\mu}{\mu - 1} \cdot mc_R, \tag{6}$$

where the markup is determined by the elasticity of substitution between orders μ. Hence, (5) and (6) highlight that, besides the demand shifters s_j, relative market sizes are determined by the marginal costs of platforms to which we turn next.

Determinants of marginal costs. The specific structure of the riders' labor market implies that marginal costs are different for platforms according to whether they hire workers as independent contractors or employees. Platforms offering C jobs pay w_C to their riders *per order* produced, which is chosen to be the numeraire of the economy and normalized to $w_C = 1$. In addition, C platforms may have to pay administrative sanctions. Consistent with these sanctions increasing with the number of missclassified riders due to size-dependent labor-law violation penalties and back-payment of taxes, we interpret these sanctions, Γ, as adding to firms' marginal costs. Thus, given that these employers do not pay social security contributions, marginal costs for C platforms are given by:

$$mc_C = w_C(1 + \Gamma) = (1 + \Gamma). \tag{7}$$

Equation (7) implies that imposing administrative sanctions will, hence, raise prices of the C platform, thereby redirecting food delivery orders towards the R platform.

Turning to the marginal costs of the R platform, its platforms pay an hourly wage $w_R(\bar{\epsilon}_R)$ plus payroll taxes τ_f. Different from C riders, R riders enjoy bargaining power through a union that bargains with the firm, taking into account the average taste parameter of its members, denoted as $\bar{\epsilon}_R$. To avoid the complication of modeling multi-worker bargaining arising from the incentives of firms to over-hire (Stole and Zwiebel, 1996; Elsby and Michaels, 2013), which is not relevant to the main point of our analysis, we assume that there is an HR division within each R platform that is disconnected from the production side and treats the contribution of each rider to the firm individually.[10] In particular, to cover labor costs, the production side of the platform transfers a fee c^{HR} to the HR division for each hour worked by a rider, which is then used to pay the corresponding gross hourly wage $(1 + \tau_f)w_R(\bar{\epsilon}_R)$. Thus, the value for this HR division of

[10] An equivalent micro-foundation for abstracting from multi-worker bargaining is to assume that the firm hires labor from intermediate firms, each one with one worker, which play the role of the indirect provider of labor services to the platforms, as in Marimon and Zilibotti (2000).

providing an additional rider to the platform becomes:

$$J_R(\bar{\epsilon}_R) \;=\; c^{HR}\bar{h}_R \;-(1+\tau_f)w_R(\bar{\epsilon}_R)\bar{h}_R \;+\; \beta(1-\delta)J_R(\bar{\epsilon}_R), \tag{8}$$

where β is the time discount factor, and δ is an (exogenous) job destruction rate. Importantly, due to the presence of search and matching frictions, the HR division incurs a flow vacancy posting cost κ when searching for a rider. Thus, the value of a vacancy is given by:

$$J_R^v(\bar{\epsilon}_R) \;=\; -\kappa \;+\; \beta\Big[\, q(\theta)J_R(\bar{\epsilon}_R) + (1-q(\theta))J_R^v(\bar{\epsilon}_R)\,\Big].$$

In equilibrium, the R platform posts vacancies until the value of the vacancy becomes zero, implying that:

$$\kappa = \beta q(\theta)J_R(\bar{\epsilon}_R). \tag{9}$$

Plugging (9) into the value of a filled job given in (8) yields:

$$c^{HR} \;= (1+\tau_f)w_R(\bar{\epsilon}_R) \;+\; \Big(1-\beta(1-\delta)\Big)\frac{\kappa}{\beta q(\theta)\bar{h}_R}. \tag{10}$$

As mentioned above, the labor costs of an R platform consist of paying a fee of $c^{HR}\bar{h}_R n_R$ to its HR division. Moreover, the production technology in (1) yields that the total number of hours worked by R riders is given by $\bar{h}_R n_R = o_R/A$. Consequently, its marginal cost of orders is given by:

$$mc_R = \frac{c^{HR}}{A}, \tag{11}$$

which highlights that, resulting from the search and matching friction, R platforms' marginal costs are driven not only by hourly wages and payroll taxes, but also by the expected discounted cost of posting vacancies.[11]

Finally, wages for R riders solve the standard Nash-bargaining problem:

$$w_R(\bar{\epsilon}_R) \;=\; \arg\max\Big\{\Big[W_R(\bar{\epsilon}_R) - U(\bar{\epsilon}_R)\Big]^{\eta}\Big[J_R(\bar{\epsilon}_R)\Big]^{1-\eta}\Big\}, \tag{12}$$

where η is the riders' bargaining weight, and $U(\bar{\epsilon}_R)$ is the outside option value of not working for the R platform (for a rider with the average taste parameter, $\bar{\epsilon}_R$.).

Summing up, differences in marginal costs and, hence, in demand for orders between the two types of platforms arise from: (i) C platforms having to pay fines after challenging

[11] Note that, unlike the marginal cost of the C platform, productivity per hour (A) enters the marginal cost of the R platform as it pays riders *per hour* worked, while the C platform pays *per order* produced.

the RL, and (ii) R platforms having to pay bargained wages with a union plus social security contributions, as well as internalizing the costly labor market search process. As discussed earlier in Section 3, the resulting bargained wage is higher in the R platform than the labor compensation in the C platforms. Hence, in the absence of administrative fines, C platforms set lower prices, which would allow them to capture a larger share of the online food delivery market.[12]

Riders' preferences There is a unit mass of workers who are ex-ante heterogeneous in their dis-utility of work, reflecting differences in caregiving responsibilities, educational commitments, and other time constraints. The taste parameter leisure, ϵ, is distributed as a left-truncated normal distribution, $\epsilon \sim N(\mu_\epsilon, \sigma_\epsilon^2) \in [0, \infty]$, across workers. Individuals are endowed with $\tilde{h}$ units of time per period, and their utility depends on both labor income, y, and the share of time allocated to leisure, $1 - h/\tilde{h}$. Specifically, the flow utility of riders is given by:

$$u(y, h) \;=\; \ln(y) + \epsilon \ln(1 - h/\tilde{h}), \quad h \in [0, \tilde{h}].$$

Modeling riders' utility as dependent on labor income captures that they are hand-to-mouth and hence do not save. Recall that riders earn labor income that depends on hours worked in R and on orders produced in C, and that the latter are not renumerated for their waiting time, leading to an hourly compensation of $\varphi A w_C$. Lastly, non-rider job seekers receive a flow value b_U from working $\bar{h}_U$ hours in an outside option representing alternative job opportunities. Therefore, consumption/income for each employment status becomes:

$$\begin{aligned}
y_U &= b_U \bar{h}_U && \text{if searcher,} \\
y_R &= w_R(\bar{\epsilon}_R)\bar{h}_R && \text{if employed in } R, \\
y_C &= (1 - \tau_c) w_C \varphi A h_C && \text{if employed in } C,
\end{aligned}$$

Accordingly, the period-by-period utility functions become:

$$u_C(\epsilon) \;=\; \max_{h_C} \ln(y_C) + \epsilon \ln(1 - h_C/\tilde{h}) \tag{13}$$

$$u_R(\epsilon) \;=\; \ln(y_R) + \epsilon \ln(1 - \bar{h}_R/\tilde{h}), \tag{14}$$

$$u_U \;=\; \ln(b_U) + \epsilon \ln(1 - \bar{h}_U/\tilde{h}). \tag{15}$$

A key simplification of our model is that the flow value of the outside option, u_U, is

[12] In our setup, note that these lower prices directly affect consumers. In practice, however, platforms may also compete in the fee they charge to restaurants. As a result, C platforms were able to sign more restaurants to their delivery services. We do not model this intermediate step. Yet, assuming that restaurants operate under perfect competition, they will pass any reduced fees to consumers in the form of lower meal prices.

exogenous due to the relatively small size of the platform sector, which we assume does not influence equilibrium outcomes in the rest of the economy.

Labor supply choices by riders. Let $U(\epsilon)$ denote the value of potential riders currently not employed as a rider (non-rider searchers, hereafter). These potential riders may be either non-employed or they may work outside of the platform sector and wish to become a rider. Hence, their value solves:

$$U(\epsilon) = u_U + \beta\Omega^U(\epsilon) \tag{16}$$

$$\Omega^U(\epsilon) = \mathbb{I}^{RC,u}_{=0}\, W_C(\epsilon) + \mathbb{I}^{RC,u}_{=1}\left[(1-p(\theta))U(\epsilon) + p(\theta)\Omega^R(\epsilon)\right] \tag{17}$$

$$\Omega^R(\epsilon) = \mathbb{I}^{R}_{=1}\, W_R(\epsilon) + \mathbb{I}^{R}_{=0}\, U(\epsilon). \tag{18}$$

The non-rider searchers decide whether to take instantly a C platform job that yields a value $W_C(\epsilon)$ or search for a job with the R platform to achieve a value $W_R(\epsilon)$. In the latter case, due to frictions, they fail to receive a job with probability $1-p(\theta)$, in which case they remain with their external option. Alternatively, with probability $p(\theta)$, they can choose between remaining with the external option or accepting the job offer in R. The policy indicator $\mathbb{I}^{RC,u}$ captures the unemployed worker's decision to search for R jobs rather than taking a C job, while $\mathbb{I}^R$ denotes their decision to accept an offer from the R platform. To reduce notation, we omit the dependence of policy functions on the taste parameter ϵ here and throughout the sequel.

When self-employed in the C platform, a rider may search for a job in the R platform. We introduce on-the-job search to capture the evidence in Figure 2 showing substantial overlap in the distributions of hours worked for R and C jobs at relatively low hours. We interpret this fact as reflecting that riders with relatively high work dis-utility would still accept C jobs while keep on searching for better-paid, shorter-hours R jobs. Then, assuming that the job offer rate, $p(\theta)$, is the same as for non-rider searchers, the resulting value becomes:

$$W_C(\epsilon) = u_C(\epsilon) + \beta\Lambda^C(\epsilon) \tag{19}$$

$$\begin{aligned}\Lambda^C(\epsilon) =& (1-p(\theta))\Lambda^{CC}(\epsilon) \\ &+ p(\theta)\left[\mathbb{I}^{CR}_{=1}\left(\mathbb{I}^{R}_{=1}W_R(\epsilon) + \mathbb{I}^{R}_{=0}\,U(\epsilon)\right) + \mathbb{I}^{CR}_{=0}\,\Lambda^{CC}(\epsilon)\right]\end{aligned} \tag{20}$$

$$\Lambda^{CC}(\epsilon) = \mathbb{I}^{C}_{=1}\, W_C(\epsilon) + \mathbb{I}^{C}_{=0}\, U(\epsilon), \tag{21}$$

where $\mathbb{I}^{CR}_{=1}$ is an indicator of the rider preferring a R job over a C job offer, while $\mathbb{I}^{C}_{=1}$ denotes that the rider prefers a C job over non-employment. Likewise, $\Lambda^{CC}(\epsilon)$ captures the value of staying with the C job or moving to the non-rider option. Note that on-the-job search in R jobs is omitted as riders could always choose to get a job directly in the

C platform. Hence, with $\Lambda^R(\epsilon)$ capturing the value of staying with the R platform or becoming a non-rider, the value for a rider in the R platform solves:

$$W_R(\epsilon) = u_R(\epsilon) + \beta\Lambda^R(\epsilon) \tag{22}$$

$$\Lambda^R(\epsilon) = (1-\delta)W_R(\epsilon) + \delta U(\epsilon), \tag{23}$$

Having characterized labor supply choices from the value of each job, we now define the two types of individuals who search for jobs in the R platform. First, there is a mass of searchers who are currently non-riders, u, and are only willing to accept a direct job offer as a rider from the R platform. Second, there is a mass of on-the-job searchers who are currently working in the C platform, c, but would prefer transiting to the R platform. Thus, the total mass of job searchers, s, is a combination of the unemployed and C riders on-the-job seekers given by:

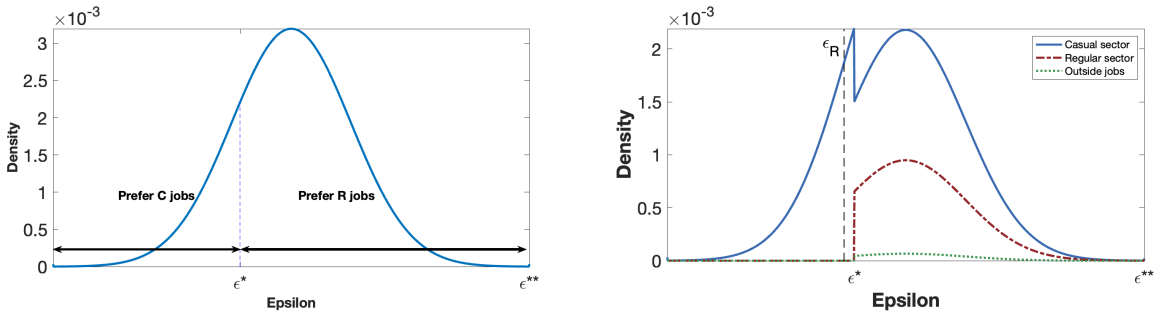
$$s = u \cdot \int_0^\infty \mathbb{I}_{=1}^{R}(\epsilon)\mathbb{I}_{=1}^{RC,u}(\epsilon)\, dG^{UR}(\epsilon) + c \cdot \int_0^\infty \mathbb{I}_{=1}^{RC}(\epsilon)\, dG^{CR}(\epsilon), \tag{24}$$

where, G^{UR} and G^{CR} are the cumulative distribution functions of the taste parameter ϵ of the u and c searchers, respectively.

Equilibrium. The equilibrium consists of prices and quantities for food delivery services (p_R, p_C, o_R, o_C), employment levels (n_R, n_C), labor supply (h_C), adjustment factor (φ), labor market tightness (θ), mean leisure taste $(\bar{\epsilon}_R)$, wage in the R platform (w_R), value functions $(J_R, J_R^v, U, \Omega^U, \Omega^R, W_C, \Lambda^C, \Lambda^{CC}, W_R, \Lambda^R)$, and distributions $(G^C, G^R, G^{UR}, G^{CR})$ such that:

- **Food delivery market:** Consumer demand (o_j, o_{-j}) maximizes utility as given by equation (5) for $j \in \{C, R\}$, while satisfying the budget constraint in (4). Moreover, platforms maximize profits and set prices according to (6).
- **Labor market:** Riders optimize labor supply h_C in C jobs as in (13). Workers' value functions satisfy (16)-(23); the value of a vacant job J_R^v satisfies the free-entry condition (9); the value of a filled vacancy J_R satisfies (8); the hourly wage w_R is set by Nash bargaining as in (12), and the factor φ adjusts the demand for and supply of hours in C jobs following (2).
- **Distributions are consistent:** $(G^C, G^R, G^{UR}, , G^{CR})$ arise from optimal policies and equilibrium outcomes, with G^R determining $\bar{\epsilon}_R$.

Figure 3: Riders' preferences for jobs and employment distribution



(a) Policy function

(b) Distribution over jobs

Source: Own elaboration from simulated data.
Note: The left panel displays workers' policies as a function of their time preferences. The right panel displays the employment distributions of the two platforms over workers' time preferences.

4.1 How the model works

Before examining in detail the strategy to quantify the model parameters, we first discuss riders' choices of their preferred job type and their resulting distribution across jobs. Panel (a) in Figure 3 shows the distribution of ϵ under the benchmark calibration for the model economy discussed in Section 5 below, along with the key cutoff value, ϵ^*, determining employment choices. Riders prefer C jobs over R jobs whenever their idiosyncratic distaste for work is low enough, namely, $\epsilon < \epsilon^*$, where ϵ^* solves $W_C(\epsilon^*) = W_R(\epsilon^*)$. The insight is that C jobs help achieve high earnings by working a high number of hours.

Conversely, riders prefer R jobs over other options whenever $\epsilon > \epsilon^*$, this time because those jobs offer them a wage premium relative to both C and outside jobs. Yet, due to search frictions, some of these riders (those without a too high taste for leisure) are willing to accept C jobs and then search on-the-job for their preferred R jobs. This happens whenever $\epsilon \in (\epsilon^*, \epsilon^{**})$, where the cutoff value ϵ^{**} solves $W_C(\epsilon^{**}) = U(\epsilon^{**})$. Finally, riders with a very high taste for leisure, $\epsilon > \epsilon^{**}$, will search for R jobs from outside jobs.

Panel (b) of Figure 3 displays the resulting employment distribution over C and R jobs in our calibrated model. It shows that 74 percent of the self-employed riders would prefer to become employees. The Figure also displays a dashed-black line that represents the level of preferences for leisure, ϵ_R, above which riders currently operating for the C platform work fewer effective hours than they would work for an R platform, $\varphi h_C(\epsilon_R) = \bar{h}$. In our calibration, this share is large: 79.9 percent of C riders work more hours than those at the R platform, but their effective hours are lower than in the latter. We note that this phenomenon arises because of the interaction of the search friction, which prevents an instant reallocation of those workers to the R platform, and the idle waiting time when working for the C platform, which reduces their effective hours. Thus, a policy that induces worker reallocation from C platforms to R jobs, *ceteris paribus*, would be

highly beneficial to these workers as they will enjoy higher earnings despite working fewer hours.

5 Model Calibration

Table 2 presents the choice of the model parameters. Besides matching moments regarding the structure of the riders' labor market, the calibration captures the riders' trade-off (see Section 3), between higher wages in R platforms and more flexible hours in C platforms. We first calibrate parameters governing the administrative sanctions, taxation, bargaining power, and some of the features of the labor market and consumers' preferences. These choices shape labor income, riders' sorting incentives into the different platforms, and the incentives of R firms to post vacancies. Next, we jointly estimate the remaining set of parameters, related to preferences and technology, using the Simulated Method of Moments (SMM) by targeting moments of hours, wages, and the relative size of the two platform types.

5.1 Calibrated parameters

The benchmark model captures the main features of the post-reform labor market. The model period is monthly, and the annual discount rate is 4.0 percent. We calibrate the administrative sanctions using annual reports from the two non-compliant firms (Glovo and Uber Eats). To measure the present discounted expected value of sanctions, we use non-compliant firms' provisions for RL-related legal risks for classifying riders as self-employed (see Appendix B). To obtain a moment comparable to our steady-state model, we compute the flow value of these provisions relative to net revenue. By 2024, expected sanctions reached 3.2 percent of net revenue, which requires $\Gamma = 0.04$.

Regarding consumers' preferences, the elasticity of substitution between orders, μ, determines the size of markups. Empirical studies quantifying market power in Spain estimate that average markups have ranged between 20 and 50 percent over the past decades (García-Perea et al., 2021). We take the lower bound of these numbers, as accounting profits are small in the food delivery sector, which leads to $\mu = 6$.

Turning to the parameters governing hours worked, we set the number of hours worked in the R platform to the mean observed in the convenience survey. Regarding the hours in the outside option, we interpret them as representing alternative opportunities to those found in the food delivery sector, including non-employment. Accordingly, we first calibrate the number of hours worked by outside job seekers as the predicted hours of workers in the Spanish economy with the same characteristics as the average rider. In particular,

Table 2: Summary of model calibration

Parameter	Description	Value	Moment
Panel I: Calibration			
A: Sanctions			
Γ	Administrative sanctions	0.04	Ratio sanctions to net revenue
B: Preferences			
$(1-\beta)^*100$	1− Discount factor	0.34	4% annual discount rate
μ	Elasticity of substitution	6	García-Perea et al. (2021)
C: Labor market			
$\bar{h}_R$	Hours worked in R	0.15	Mean hours in $R/24$
$\log(\kappa)$	Vacancy costs	7.04	Hagedorn and Manovskii (2008)
η	Workers' bargaining weight	0.50	Petrongolo and Pissarides (2001)
δ	Destruction rate in R	0.07	EU flow $R = 7\%$
D: Outside option			
$\bar{h}_U$	Hours outside option	0.16	Predicted hours worked/24
$\log(b_U)$	Flow value outside option	0.89	Marginal C rider is indifferent
E: Taxes			
τ_c	Payroll taxes in C	0.16	Payroll taxes self-employed
τ_f	Payroll taxes in R	0.29	Payroll taxes employers
Panel II: SMM			
A: Preferences			
s_R/s_C	Utility shifter goods	1.70	Employment share of R platform
μ_ϵ	Work disutility: Mean	3.60	Mean hours in $C/24 = 0.22$
σ_ϵ	Work disutility: Std. dev.	0.62	95th percentile hours in $C = 7/24$
B: Technology			
$\log(A)$	Hourly productivity	2.54	Mean log wages in $R = 2.07$
ι	Matching parameter	0.23	Wage premium in $R = 0.17$

Note: This Table describes the quantification of the parameters and their respective targets.

we regress the annual hours worked by an individual i, h_i, on a vector of characteristics, X_i, including sex, age, nationality, skills, occupation, and sector. To estimate this regression, we use annual microdata from the 2022 wave of the Wage Structure Survey (EES), which is the only wave available with information on hours worked for a representative sample of employees in Spain during our reference period. To determine the average sociodemographic characteristics of riders, $\bar{X}_i$, we draw data from the Adigital (2020) survey. Using the OLS estimated parameters, $\hat{\beta}$, from that regression yields a prediction of $\hat{\beta}\bar{X}_i = 5.0$ hours worked for workers with the average characteristics of riders in similar occupations. We then adjust this number for the unemployment rate among such workers using quarterly data from the Spanish Labor Force Survey (LFS) for the last two decades. Our main finding is that the predicted unemployment probability for this specific type of worker is 24 percent. Assigning zero hours when unemployed, we obtain

$\bar{h}_U = 0.76 \cdot 5.0/24 = 0.16$. Since we set the fraction of hours worked in the R platform equal to 0.15, consistent with a mean daily hours of 3.7, our calibration strategy implies that riders expect to work similar hours in outside jobs and with the R platform.

Next, we discuss the choice of those parameters guiding labor market flows in the R platform. Following Hagedorn and Manovskii (2008), the flow vacancy cost κ is set to match an expenditure made by firms equivalent to forgoing 58 percent of average labor productivity per worker. As is common in search and matching models (Petrongolo and Pissarides, 2001), the bargaining power of riders, η, is set equal to 0.50. Both values are in the ballpark of what is often used to calibrate frictional models of the Spanish labor market (see, e.g., Bentolila et al., 2012). Regarding employment separations, we choose the monthly job destruction rate, δ, to match a 7.0 percent transition rate from employment to non-employment in the Postal Services and Courier Activities sector (NACE 532H) of the largest Spanish administrative dataset (Muestra Continua de Vidas Laborales-MCVL) for the post-reform period August 2021-December 2022.[13]

As for payroll taxation, we choose tax rates in the C and R platforms to match the average payroll taxes of the self-employed, $\tau_c = 16$ percent, and the employers, $\tau_f = 29$ percent, respectively.

5.2 Jointly estimated parameters

The remaining six parameters $\mathbf{o} = (s_R, \mu_\epsilon, \sigma_\epsilon, \iota, A, b_u)$ are related to consumers' preferences, riders' tastes for hours worked, the matching and production technologies, and the flow value of riders' outside option. These parameters, which jointly determine the distributions of market shares of two types of platforms, hours worked, wages, and employment rates, are estimated using an SMM approach. Below, we describe each parameter and its closest associated moment. Moreover, Table A.1 in Appendix A reports the fit of the targeted moments, which turns out to be very satisfactory.

The utility shifter s_R is informative about the relative size of the R platform by affecting the relative demand for food delivery services, as shown in Equation (5). Therefore, following the results from the convenience survey in Table 1, we include 24 percent as the target for the employment share of riders in these platforms.

The distribution of work hours in the C platform is informative about riders' work tastes, ϵ, since self-employed riders choose their own working hours. Note that the model controls for the self-selection of workers with a strong taste for flexible hours into the C

[13]This NACE sector includes riders together with postal staff and other types of couriers. Since these workers have more stable labor contracts than riders, we restrict the sample to employees who work for two consecutive months and whose tenure is no longer than 18 months, to be consistent with the tenure reported in Table 1.

platform. In the data, we observe an average of 5.4 and a 95th percentile of 7.0 hours worked in the C platform. We use these moments to estimate the mean, μ_ϵ, and standard deviation, σ_ϵ, of idiosyncratic preferences.

Next, consider the parameters related to production and matching technologies. Given the exogenous consumer's spending, the productivity per hour, A, determines the marginal revenue productivity of riders and thereby affects hourly wages in both platforms. By influencing the matching probabilities, the parameter ι determines the size of the match surplus for R jobs, which influences the wage premium in the R relative to the C platform. To discipline these parameters, we target an average hourly wage net of social security contributions of €7.9 euros in the R platform, based on the sector's collective bargaining agreement.[14] In addition, using the evidence from the survey, we also target a wage premium of 17.2 log points for R jobs.

Lastly, we turn to the flow value of outside jobs, b_U. This parameter reflects the expected flow value of riders outside the delivery sector (including non-employment). As we calibrate the model to cross-sectional data with information on workers who work as riders, i.e., choose working in the food delivery sector over the outside option, we target that the marginal worker is indifferent between both alternatives, $W_C(\epsilon^{**}) = U(\epsilon^{**})$. As a result, the only non-employment in our benchmark calibration arises from frictional non-employment, i.e., those riders who lose their R jobs.

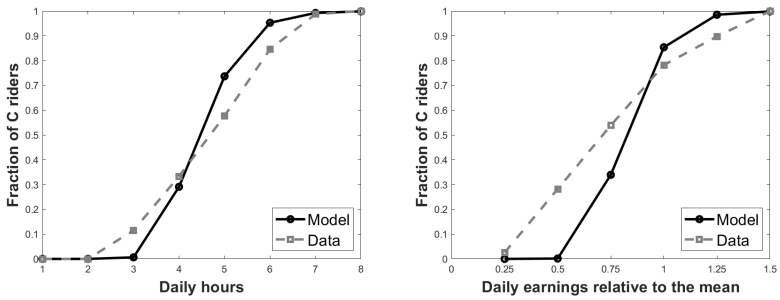
5.3 Model fit of untargeted moments

To validate the predictions of the model, we compare how the model fits untargeted moments related to hours worked and earnings of C riders after the RL.

A key feature of the data is that the hours distribution in C jobs is smooth and exhibits relatively high dispersion, which our calibration interprets as reflecting dispersed tastes for working hours among its riders. Reproducing these features of the data is policy-relevant because they reflect the value that workers place on the flexibility of work schedules. Figure 4(a) displays the cumulative distribution of daily hours worked among C workers in the data and model after the RL. The distribution in the model closely aligns with the observed distribution in the data. Specifically, the model reproduces that about one-third of C riders work at most 4 hours per day, while nearly one-tenth exceed 6 daily hours, and the majority work around 5 hours.

[14] We use the collective agreement (M/441/2021/N), signed between Just Eat and trade unions in 2021 as a representative source of information for remuneration of R riders. We compute our estimate of hourly wages by dividing the annual base wage (€15,232), net of employees' social security and income taxes, by the annual number of hours for full-time riders (1,792). We find net hourly wages to be €7.8. This result is reassuring, as we find that net hourly wages of R riders in the convenience survey are around €7.0

Figure 4: Model fit of hours worked and earnings in C jobs



(a) Hours worked (b) Earnings

Note: The left panel displays the cumulative distribution of hours worked among workers in C jobs in the data and model, and the right panel displays the cumulative distribution of daily earnings (relative to the mean) among C workers in the data and model. The model and data periods are post-Rider's Law.

Another policy-relevant feature is the varying importance of C jobs in terms of earnings for riders. Figure 4(b) displays their post-reform cumulative distribution among C riders both in the data and the model, where earnings are expressed relative to the mean. Two facts stand out. First, the model replicates earnings dispersion across riders well. Since it abstracts from hourly wage dispersion, this result indicates that dispersion in hours worked alone is sufficient to rationalize earnings inequality among riders. Second, the model somewhat underestimates the share of riders with relatively low earnings; nevertheless, it correctly replicates that most riders earn within 25 percent of the mean.

6 Policy experiments

This section quantifies the effects of the RL on the online food delivery sector by comparing our baseline (post-reform) economy ($\Gamma = 0.04$) with a counterfactual economy where hiring self-employed riders does not carry administrative sanctions with it ($\Gamma = 0$). The mechanism of how sanctions affect the structure of the food delivery market is explicit in our model: compared to the counterfactual economy without sanctions, sanctions in the benchmark economy increase the marginal cost of using C riders, which in turn raise the relative prices of C orders, p_C/p_R, and shift consumer demand to R platforms, with a subsequent labor demand change across platforms.

6.1 Labor-market effects of the Rider's Law

Before turning to the main counterfactual results, a few clarifications about the correct interpretation of these findings are in order. First, we abstract from the fiscal implications of the reform. With the resulting fines and changes in payroll-tax revenues being absorbed by the general treasury, the effects of the RL should be interpreted as exclusively affecting

Table 3: The effects of the RL on the food delivery sector

	No sanctions (Γ=0)	Baseline (Γ=0.04)		No sanctions (Γ=0)	Baseline (Γ=0.04)
A: Orders			*D: Employment*		
Relative price (p_C/p_R)	0.52	0.54	Employment C	0.79	0.74
Market share (o_C/o_R)	0.79	0.73	Employment R	0.20	0.24
Adjustment factor (φ)	0.68	0.63	Outside jobs	0.01	0.02
B: Wages			*E: Hours*		
Mean log hourly wages C	1.95	1.87	Mean effective hours C	3.65	3.35
Mean log hourly wages R	2.10	2.04	Mean effective hours	3.66	3.45
Labor market tightness	0.39	0.41			
C: Job preferences			*F: Welfare change (%)*		
Cutoff C and R jobs, ϵ^*	3.29	3.06	Consumers		-1.98
Cutoff C and U jobs, ϵ^{**}	5.99	5.99	Riders		-7.24

Note: This Table displays the model results from the counterfactual simulation where defier, C, platforms do not have to pay a fine: $\Gamma = 0$.

this specific sector of the economy. Second, as stressed above, we also abstract from sector-wide demand trends, implying that our results should be interpreted as a counterfactual of how the riders' sector would have fared absent the RL with those trends taken as given.

Panel A in Table 3 shows that administrative sanctions directly translate into a convergence of relative prices of orders, as C platforms raise prices to offset the increase in marginal costs and maintain profitability at the margin. We find that the relative price of C orders is 5.6 percent higher in the benchmark (p_C/p_R =0.54) relative to the counterfactual without sanctions (p_C/p_R =0.52). Though there is no detailed data on prices charged to customers for identical food deliveries, there is media evidence that Glovo (C) significantly charged higher commissions in the post-reform relative to the pre-reform period, while those of Just Eat (R) remained at similar levels (see, e.g. Qamarero, 2024).

Note that a higher relative price of C orders shifts consumer demand to R platforms. In particular, we find that the RL increases the output share of the latter platform by 5.9 percentage points, which aligns exactly with the evolution of market shares in the data after the implementation of the reform (see Figure 1).

As consumers demand fewer orders from the C platform after the RL, C riders face longer waiting times. The magnitude of this increase depends on how large the reduction in working hours on the C platform is relative to its decline in orders. The discussion in Section 4.1 highlights that this trade-off is unfavorable, as many riders leaving the C platform produce more orders in the R platform than they previously did with the C

platform. As sanctions lead to a decline in the total number of orders, it must be that the remaining C riders produce fewer orders per hour worked – i.e., waiting times increase.[15] Consequently, φ declines markedly from 0.68 to 0.63. We are not aware of any quantitative evidence that would allow us to test this prediction directly. However, qualitatively, there is evidence about riders reporting longer waiting times for their account to be activated by Glovo after the RL (see, e.g., Rodríguez Eiris, 2024).

Panel B in Table 3 shows that, as waiting times increase on the C platform, the average net hourly wage of its riders decreases by 8.1 log points relative to a counterfactual without sanctions. The reduction in C wages implies that sanctions induce a fraction of long-hours C riders to prefer R jobs, as illustrated by the decrease in the cutoff value ϵ^* in the baseline economy relative to the economy without sanctions (panel C). Overall, this leads to a fall of 4.4 percentage points in the employment share of the C platform (panel D), which is about 5.6 percent lower than its corresponding share with no sanctions. At the sector average level, sanctions reduce average effective hours worked (panel E). This is due to riders' reallocation from the long-hours (C) to the short-hours (R) platform, as well as to the increase in waiting times for riders staying in the C platform.

The negative effects of sanctions on the C platform have meaningful spillover effects on the R platform. On the one hand, since working in a C platform becomes less attractive and the conditions of non-rider jobs are not affected by the RL, R riders' outside option worsens, weakening their bargaining position and pushing R wages down. On the other hand, the rise in the demand for R orders leads to a higher price, p_R, which is partially passed through to higher wages as their marginal revenue product of labor increases. Taken together, we find that the worsening of the outside option is the dominating channel, leading to a decrease of 6.0 log points in the R wage compared with the economy lacking sanctions. As a result, the wage premium does not change much due to the presence of sanctions. Although there is not microdata on wages available for the pre-reform period without sanctions, there is some aggregate information about net hourly wages from the main C and R platforms. In 2019, Glovo paid their self-employed riders between €3.5 to €4 while Just Eat paid €5, once social security contributions and other expenses are deducted. Thus, these data suggest that the wage premium in favour of R riders was also around 20 percent before the RL (see, e.g., El País, 2020).

Higher demand for R orders boosts vacancy creation and raises labor market tightness, θ, increasing the R employment share by 4.1 percentage points, from 0.20 to 0.24 (panel B). In other words, without sanctions in place, one-fifth of employee riders would have been self-employed instead. Overall, the R platform absorbs nearly all of the employment

[15] In principle, total employment could decrease enough for the opposite to happen. Yet, in our simulations, this effect is not sufficiently strong.

decline in the C platform due to sanctions, with the overall sector contracting by 1 percentage point. Put differently, the effects of the RL on total employment are small.

6.2 Welfare effects of the Riders' Law

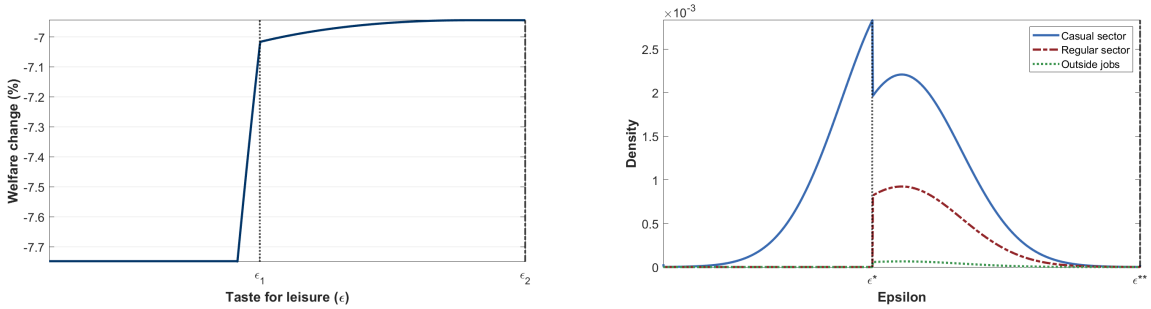
The RL affects welfare through changes in household consumption and riders' labor income and hours worked. Welfare is measured in terms of consumption-equivalent variation, that is, the percentage of lifetime consumption that an individual would be willing to forgo to remain indifferent between the benchmark economy with sanctions and the counterfactual without them.

As reported above, the RL contracts the food delivery sector, thereby reducing consumer demand in comparison to the counterfactual economy. Moreover, as prices from C platforms rise more than prices from the R platform, there is a reallocation of consumption across imperfectly substitutable orders. Consequently, we find that sanctions reduce consumer welfare from food delivery by 1.8 percent.

On the flip side, rising prices have the potential to increase riders' welfare, as the sector as a whole produces fewer orders and, hence, requires fewer effective hours for the same total income, m. In addition, as shown in Section 4.1, many workers with C jobs would prefer to transit to the R platform. In fact, in partial equilibrium, the reallocation towards employees would be highly beneficial, as it allows these workers to work fewer hours at R jobs while receiving higher pay than at the former C jobs. Nevertheless, in general equilibrium, we find that the RL leads to an average-rider welfare loss of 7.2 percent.

These welfare losses depend on riders' idiosyncratic preferences over working hours, as shown in the left panel of Figure 5. Specifically, we define two groups of riders based on the following cutoff values: (i) ϵ^*, such that $W_R^{\text{NoRL}}(\epsilon^*) = W_C^{\text{NoRL}}(\epsilon^*)$, where the superscript "NoRL" in the riders' value functions indicates that the economy operates without sanctions for violating the RL. The figure shows that nearly one-third of riders always prefer the C platform due to their low disutility from hours worked ($\epsilon < \epsilon^*$). These riders suffer the largest welfare losses due to the substantial pay reduction in the C platform, without enjoying any offsetting gains from improved job opportunities in the R platform. The remaining two-thirds of riders, with $\epsilon \in (\epsilon^*, \epsilon^{**})$, would prefer to work for the R platform. Among them, those currently with the C platform engage in on-the-job search. Two opposing effects arise. On the one hand, these riders benefit from the RL, as it facilitates transitions from C jobs to R jobs. On the other hand, they experience earnings and consumption losses because: (i) C platform wages decline, hurting them while searching for an R job, and (ii) R platform wages also decline due to

Figure 5: Welfare effects of the Riders' Law



(a) Welfare change

(b) Distribution over jobs without sanctions

Note: The left panel of this Figure displays the welfare effect of the Riders' Law relative to the economy without sanctions against workers' taste for leisure. The right panel shows the distribution of riders over the different types of jobs in the counterfactual scenario without sanctions.

a lower option value in the C platform. Overall, the negative effects outweigh the positive ones, leading to a decline in riders' overall welfare.

A relevant question is whether the same reform would have increased welfare if the riders' market had been structured differently. We find that the answer is affirmative when riders' waiting time in the C platform does not increase excessively, so that their hourly wage does not fall by much. A substantially higher flow value of outside jobs ($\ln(b_U) = 1.79$) would achieve exactly this.[16] We provide details on this thought experiment in Appendix C and only briefly sketch the intuition here. A high outside option implies that relatively few workers are employed in the C platform with effective hours below those in the R platform before the reform ($\varphi h_C < \bar{h}_R$), as riders with a high preference for leisure choose outside jobs rather than working in the C platform. As discussed earlier, having only few unproductive workers in the C platform prior to the reform contributes to a small increase in waiting time when sanctions are in place. The data, however, reject this alternative view of the market: it would imply no overlap in hours worked across platforms, since there would be no C riders working few hours.

In addition, a comment on the potential effects of sanctions on the size of the food delivery sector is worth mentioning. Given that we observe a cross-section of riders in the post-reform period, our calibration implies that all workers within the model prefer to be riders over outside jobs, even with sanctions in place. Hence, our model abstracts from the possibility that, in the absence of sanctions, food delivery could expand by attracting workers from other sectors. However, incorporating this margin would strengthen our results on welfare, as the sanctions would further shrink the food delivery sector.

Lastly, recall from Table 1 that about 18.0 percent of riders lack work permits. One

[16] In this thought experiment, we simultaneously decrease search frictions to keep the wage premium of the R platform unchanged.

may argue that politicians are less concerned about their employment prospects because these workers cannot vote in elections. As these riders work particularly long hours, i.e., they are those endowed with low ϵ, their weight in our welfare calculations is relatively high. However, even ignoring the 18 percent of workers with the lowest ϵ from our welfare calculations, the RL still leads to a 7.1 percent welfare loss among riders.

6.3 Complementing the Riders' Law with transfers

So far, we assume that any fiscal effects of the reform, including the sanctions, are absorbed by the general treasury. While this assumption is consistent with observed practice, one may also ask whether the reform would be less detrimental if the revenue from the sanctions were redistributed to riders. We find that the welfare effects of such a transfer depend on its specific design. A lump-sum transfer to all riders significantly reduces welfare losses. In contrast, subsidizing only riders on the R platform through a reduction in payroll taxes further exacerbates welfare losses.

6.3.1 Lump-sum transfers

We implement a lump-sum transfer T that satisfies the budget balance constraint, $T = o_C \cdot \Gamma$. This transfer reduces the welfare loss for the average rider from 7.2 percent to 2.5 percent. Put differently, most of the welfare loss stems from the absence of rider compensation during the reform. Fully compensating riders for the reform would require a transfer exceeding sanction revenue by 50.0 percent ($T = 1.5 \cdot o_C \cdot \Gamma$).[17]

Two additional issues are worth noting. First, lump-sum transfers do not alter decisions by firms and workers, so the food delivery sector still shrinks and consumer welfare still declines. Second, even a transfer that preserves average welfare does not avoid that riders remaining on the defying C platform are worse off due to lower wages (see Figure A.1 in Appendix A).

6.3.2 Reducing payroll taxes

Instead of a lump-sum transfer, perhaps a more natural policy for the government may be to complement the RL with lower social security contributions for R riders, as such a reform is more directly targeted at the jobs it intended to strengthen. We study a reduction in the employer-paid tax rate in R from 29 to 16 percent. This cut decreases marginal costs of the R platform and, thus, leads to a 3.4 percent decline in its prices. To meet the additional demand, R firms increase vacancy creation, driving labor market tightness and wages up by 10.3 percent and 0.3 percent, respectively.

[17] As the sector is small overall, the resulting deficit (€13.3 million) represents just 0.002 percent of total Spanish public revenues.

Nevertheless, riders' average welfare *decreases* by 1.0 percent. The reason for this seemingly puzzling result is closely related to why the RL reduces welfare; that is, similar to sanctions in the RL, the payroll tax bonus shifts demand from the C to the R platform. However, as the marginal rider switching from the C to the R platform completes more efficient hours at the latter, idle waiting time of the remaining workers in the C platform must increase.[18] In turn, the earnings of C workers go down, as does their welfare. Finally, the worsening conditions on the C platform deteriorate R riders' bargaining position, which explains the modest increase in their wages despite a higher labor market tightness. In sum, this policy is less effective than lump-sum rebates in terms of welfare.

7 Conclusions

This paper quantifies the impact on employment, hours worked and wages of policies mandating the use of dependent employees by online food delivery platforms. We focus on Spain, which is a forerunner in implementing such policies through the so-called Riders' Law, enacted in 2021. This reform established the presumption of dependent employment for workers in this sector, known as riders, and levies large fines on platforms not complying with this legislation. Using information on non-compliant platforms' provisions for RL-related legal risks, we estimate that sanctions account for 3.2 percent of their net revenue. A counterfactual simulation shows that nearly one-fifth of the employee riders after the reform would have been self-employed in the absence of sanctions. Hence, the reform significantly incentivizes employee status in the food delivery sector. However, sanctions lead to an average welfare loss among riders of 7.2 percent due to lower wages, especially among those who remain working as bogus self-employed for the defier platforms. Most of this welfare loss stems from longer waiting time by C riders, which is not paid by their platforms. Accordingly, a lump-sum rebate to riders of the amount of revenue raised by sanctions decreases the welfare loss of the average rider to 2.5 percent, while incurring an extra deficit of 50 percent relative to the sanctions would fully compensate the average rider relative to the economy without sanctions.

To reach these conclusions, we develop a quantitative equilibrium model with heterogeneous workers who sort into self-employed and employee jobs, each offered by a different platform that provides services to a representative consumer. These jobs are heterogeneous in terms of employability, work time flexibility, idle working time, and wages, capturing the main trade offs in the policy debate. The model is calibrated using several data sources, including a convenience survey, platforms' annual reports, and aggregate statistics from the food delivery sector. This calibration implies that riders

[18] In principle, total demand could grow enough for the opposite to happen. Yet, in our simulations, this effect is not sufficiently strong.

with relatively low tastes for leisure choose C jobs because they offer greater flexibility in working long hours and are instantaneously available. In contrast, the remaining riders prefer to work in the R sector, due to its wage premium, though they often accept C jobs while seeking R jobs to avoid non-employment or less preferable outside jobs.

Our finding that fines imposed on non-complying platforms are partially passed through to workers via lower employment, wages, and ultimately welfare should not be surprising. Yet, our model allows for two equilibrium mechanisms softening the blow to riders' labor outcomes, which, in principle, could be welfare-enhancing. First, resulting from higher wages and fewer hours worked at R jobs, shifting employment to these jobs improves, ceteris paribus, riders' welfare. Second, as fines affect the marginal costs of platforms, firms shift part of the burden to consumers/restaurants through higher prices, which could also benefit riders. We show that a key reason for the decrease in welfare is the increase in idle working time at C jobs. This not only adversely affects workers remaining at that platform but also decreases the bargained wage at the R platform as riders' outside option deteriorates. Longer waiting time, in turn, is a direct implication of some riders' having very low effective working time at C jobs, which is a phenomenon well-supported by the data.

Finally, given that we highlight differences in wages, hours flexibility, and search frictions between the two types of jobs, we see several promising avenues for future research, including extensions of the model that include other regulatory differences between the two sectors, such as overtime pay, maximum hours regulations, vacation days, and health protection benefits.

References

Adigital (2020). Importancia económica de las plataformas digitales de delivery y perfil de los repartidores en España. Technical report, Asociación Española de la Economía Digital (Adigital).

Albrecht, J., Navarro, L., and Vroman, S. (2009). The effects of labour market policies in an economy with an informal sector. *The Economic Journal*, 119(539):1105–1129.

Angrist, J. D., Caldwell, S., and Hall, J. V. (2021). Uber versus taxi: A driver's eye view. *American Economic Journal: Applied Economics*, 13(3):272–308.

Bentolila, S., Cahuc, P., Dolado, J. J., and Barbanchon, T. L. (2012). Two-tier labour markets in the great recession: France versus Spain. *The Economic Journal*, 122(562):155–187.

Boeri, T., Giupponi, G., Krueger, A. B., and Machin, S. (2020). Solo self-employment and alternative work arrangements: A cross-country perspective on the changing composition of jobs. *Journal of Economic Perspectives*, 34(1):170–195.

Borja, I. (2026). Uber Eats: 110 millones por 60.000 repartidores falsos autónomos. *Alerta Trabajo.*

Bracha, A. and Burke, M. A. (2021). How big is the gig? the extensive margin, the intensive margin, and the hidden margin. *Labour Economics*, 69:101974.

Carry, P. (2022). The effects of the legal minimum working time on workers, firms and the labor market. *Manuscript, Princeton University.*

Casado, O. (2025). Así evade Uber la mayoría de sus multas. *Economia Digital.*

Chen, M. K., Rossi, P. E., Chevalier, J. A., and Oehlsen, E. (2019). The value of flexible work: Evidence from Uber drivers. *Journal of Political Economy*, 127(6):2735–2794.

Cooper, R., Meyer, M., and Schott, I. (2017). The employment and output effects of short-time work in Germany. Technical report, National Bureau of Economic Research.

Datta, N., Rong, C., Singh, S., Stinshoff, C., Iacob, N., Nigatu, N. S., Nxumalo, M., and Klimaviciute, L. (2023). Working without borders: The promise and peril of online gig work. Technical report, The World Bank.

Delivery Hero SE (2022). Annual report 2021. Technical report.

Delivery Hero SE (2023). Annual report 2022. Technical report.

Delivery Hero SE (2024). Annual report 2023. Technical report.

Delivery Hero SE (2025). Annual report 2024. Technical report.

Den Haan, W. J., Ramey, G., and Watson, J. (2000). Job destruction and propagation of shocks. *American Economic Review*, 90(3):482–498.

Dolado, J. J., Lalé, E., and Turon, H. (2025). Zero-hours contracts in a frictional labor market. *SSRN 3988202, The Economic Journal (forthcoming).*

El Español (2021). Uber eats, just eat, glovo y deliveroo: así funciona el reparto a domicilio.

El País (2020). Los repartidores también sufren la crisis: ¿cuánto gana un rider de Glovo, Uber Eats, Delveroo y Just Eat?

Elsby, M. W. L. and Michaels, R. (2013). Marginal jobs, heterogeneous firms, and unemployment flows. *American Economic Journal: Macroeconomics*, 5(1):1–48.

Feng, Y., Lagakos, D., and Rauch, J. E. (2024). Unemployment and development. *The Economic Journal*, 134(658):614–647.

Frazier, N. (2018). *An equilibrium model of wage and hours determination: Labor market regulation in the retail sector*. PhD thesis, Rice University.

García-Perea, P., Lacuesta, A., and Roldan-Blanco, P. (2021). Markups and cost structure: Small spanish firms during the great recession. *Journal of Economic Behavior & Organization*, 192:137–158.

Garin, A., Jackson, E., and Koustas, D. (2022). Is gig work changing the labor market? key lessons from tax data. *National Tax Journal*, 75(4):791–816.

Garin, A., Jackson, E., Koustas, D. K., and Miller, A. (2023). The evolution of platform gig work, 2012-2021. Technical report, National Bureau of Economic Research.

Haanwinckel, D. and Soares, R. R. (2021). Workforce composition, productivity, and labour regulations in a compensating differentials theory of informality. *The Review of Economic Studies*, 88(6):2970–3010.

Hagedorn, M. and Manovskii, I. (2008). The cyclical behavior of equilibrium unemployment and vacancies revisited. *American Economic Review*, 98(4):1692–1706.

ILO (2024). Realizing decent work in the platform economy. © International Labour Organization.

Katz, L. F. and Krueger, A. B. (2019). The rise and nature of alternative work arrangements in the United States, 1995–2015. *ILR review*, 72(2):382–416.

Marimon, R. and Zilibotti, F. (2000). Employment and distributional effects of restricting working time. *European Economic Review*, 44(7):1291–1326.

Mas, A. and Pallais, A. (2017). Valuing alternative work arrangements. *American Economic Review*, 107(12):3722–3759.

Petrongolo, B. and Pissarides, C. A. (2001). Looking into the black box: A survey of the matching function. *Journal of Economic Literature*, 39(2):390–431.

Portier Eats Spain (2025). Annual report 2024. Technical report.

Público (2023). Trabajo impone a Glovo una nueva sanción por importe de 56,7 millones de euros. *Público*.

Qamarero (2024). Comisiones delivery: ¿cuánto cobran realmente las plataformas? Accessed: April 2, 2026.

Rodríguez Eiris, E. (2024). Glovo desespera a los riders al tardar «semanas o meses» para activar las nuevas cuentas. Accessed: April 2, 2026.

Satchi, M. and Temple, J. (2009). Labor markets and productivity in developing countries. *Review of Economic Dynamics*, 12(1):183–204.

Scarfe, R. (2019). Flexibility or certainty? The aggregate effects of casual jobs on labour markets. *Edinburgh School of Economics Discussion Paper*, 294.

Stanton, C. T. and Thomas, C. (2025). Who benefits from online gig economy platforms? *American Economic Review*, 115(6):1857–1895.

Stole, L. A. and Zwiebel, J. (1996). Intra-firm bargaining under non-binding contracts. *The Review of Economic Studies*, 63(3):375–410.

Zenou, Y. (2008). Job search and mobility in developing countries. theory and policy implications. *Journal of Development Eonomics*, 86(2):336–355.

Appendix

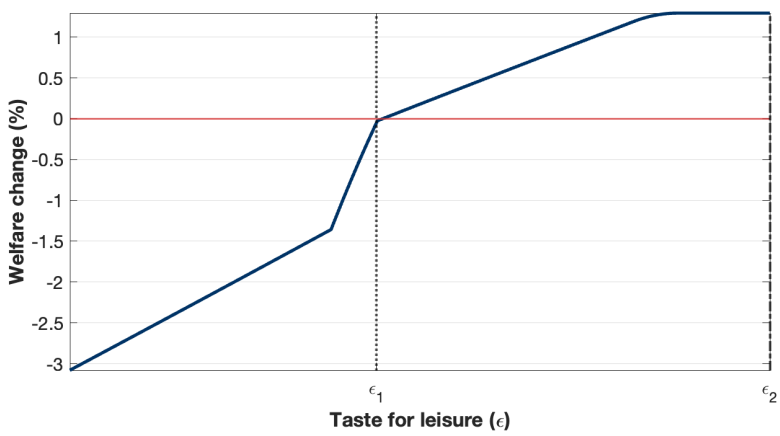
A Additional figures and tables

Table A.1: Model fit of estimated parameters

Parameter	Moment	Model	Data
A: Preferences			
s_R/s_C	Employment share of R platform	0.24	0.24
μ_ϵ	Mean hours in $C/24 = 0.22$	5.47	5.39
σ_ϵ	95th percentile hours in $C = 7/24$	6.96	7.00
B: Technology			
A	Mean log wages in R	2.04	2.07
ι	Wage premium in R	0.17	0.17

Note: The table reports the values of the SMM approach targeted moments in the model and the data.

Figure A.1: Welfare effects of the RL with welfare-neutral lump-sum transfers



Source: Own elaboration from simulated data.
Note: This Figure displays the welfare effect of the RL ($\Gamma = 0.04$) and the welfare-neutral transfer ($T = 1.5 \cdot \Gamma o_C$) as a function of workers' time preferences (ϵ).

B Estimating the size of government sanctions

This section documents the data sources and calculations underlying the estimation of the administrative sanctions from the RL. We use data from Glovo and Uber Eats, the two defier platforms that remained in the Spanish food delivery market after the RL.

Revenue of the C platforms in Spain (2021–2023). Table B.1 summarizes the resulting estimates of Glovo's revenue in Spain. The 2021 figure (€203.8 million) cor-

responds to the consolidated revenue of Glovoapp Spain Platform S.L. as filed with the Commercial Registry (*Registro Mercantil*). For 2022 and 2023, following Glovo's acquisition by Delivery Hero, we estimate Glovo Spain's revenue as a share of Delivery Hero's published consolidated net revenues for the European region (€980.5 million in 2022 and €1.522.4 million in 2023), consistent with Glovo Spain's weight of 36 percent in terms of net revenue in 2021 (Delivery Hero SE, 2022, 2023, 2024). We estimate a lower and upper bound based on a deviation of 10 percentage points with respect to this baseline number.

Administrative sanctions (2021–2024). Our analysis focus on the amount of administrative sanctions that influence the decision-making of defiant firms through affecting their operating costs. Two facts are especially relevant for our purposes. First, these sanctions increase platforms' variable costs. Specifically, the Labour Inspectorate and Social Security (*Inspección de Trabajo y Seguridad Social*, ITSS hereafter) sanctioned Glovo for subcontracting 37,348 riders as independent contractors, reaching €205.3 million in total as a result of the sum of a €123.5 million from a penalty for the labour-law violation (*acta de infracción*) and €80.0 million sanction from back-payment of avoided Social Security contributions (*acta de liquidación*) (Público, 2023). Since Uber Eats subcontracted fewer riders, this company accumulated only €23.2 million in sanctions from the ITSS during the same period, comprising a penalty for labour-law violations (€19.75 million) and back-payment of Social Security contributions (€3.45 million).

Second, the sanctions by the ITSS do not capture the present discounted expected value of sanctions faced by both platforms, which is the object of interest in our analysis. The reasons are that: (i) platforms could accumulate even more sanctions in the future, and (ii) the Spain's National Court (*Audiencia Nacional*) did not yet confirmed the sanctions during the reference period (Casado, 2025).[19]

To measure the present discounted expected value of sanctions, we turn to the non-compliant firms' provisions for RL-related legal risks for classifying riders as self-employed. Table B.1 reports the amount of these provisions by December 2024 according to their annual reports. In total, defiant platforms accumulated €923.5 million in RL-related provisions for legal risks up to December 31, 2024.

Estimate of sanction-to-revenues. Since our model is in steady-state, we compute the flow value of these sanctions using a discount factor of 0.96, implying a total flow sanction of €26.5 million among non-compliant platforms. To obtain a comparable moment

[19]For instance, the first sanctions against Uber Eats were not upheld by the courts until March 2026 (Borja, 2026).

Table B.1: Provisions for RL-related legal risks and net revenue (2024, million of euros)

	Glovo	Uber Eats
Provisions	492.2	171.3
Net revenue	681.1	136.1

Note: For Uber Eats, we use consolidated net revenue (*importe neto cifra de negocios*) reported by Portier Eats Spain S.L. in its annual accounts, which is the legal entity through which Uber Eats operates in Spain (Portier Eats Spain, 2025). For Glovo, we combine figures from annual reports with estimates. For 2021, we use consolidated net revenue reported by Glovoapp Spain Platform S.L. After this year, Glovo merged into the Delivery Hero group. Thus, for 2024, we estimate Glovo Spain's revenue as a share of Delivery Hero Group's European segment revenue, as reported in its Annual Reports (Delivery Hero SE, 2025). This share (36%) is derived from the ratio of Glovoapp SA's revenue to Delivery Hero's European segment revenue in 2021 (Delivery Hero SE, 2022), the last year prior to Delivery Hero's acquisition of Glovo; specifically, $203.8/571.4 \approx 0.36$. Regarding the source of provisions, Delivery Hero SE (2025) establishes "Delivery Hero has provisioned €492.2 million for legal risk related to rider reclassification and connected VAT risks". Portier Eats Spain (2025) increased its provisions from 0 by December 2020 to €136.3 million by December 2021, which further increased to €171.3 million by December 2024.

within the model, we express the flow value of these provisions relative to non-compliant platforms' net revenue as of 2024, which yields:

$$\frac{\text{Flow value of sanctions}}{\text{Net revenue}} = \frac{26.5}{817.2} = 0.032.$$

Thus, we estimate that expected sanctions reach 3.2 percent of net revenue for C platforms. This is the target in our calibration exercise, leading to $\Gamma = 0.04$ in the model. Note that this likely represents a lower bound for the underlying sanctions of the reform, as Glovo Spain likely represents a smaller fraction of Delivery Hero's net revenue than the one we use in our estimates (e.g., this group has acquired more firms over time).

C Understanding the welfare effects of the RL

This section discusses how our model allows for the possibility that the RL could be welfare-improving. To do so, we parameterize an alternative model economy in which the RL is beneficial and then show that the data lead us to reject this parameterization.

We focus on the minimum parameter adjustments required for the RL to yield a welfare gain slightly above zero for the average rider. A key condition for this to happen is that the wage in the C platform does not deteriorate much, namely, idle working time does not increase excessively. For that to be the case, the technological constraint in Equation (2) establishes that

$$d\log(\varphi) \approx 0 \iff d\log(o_C) \approx d\log\Big(n_C \cdot \mathbb{E}(h_C(\epsilon)\Big).$$

In other words, the percent decline in orders must be similar to the percent decline in total hours worked following the RL. To achieve this, our alternative market structure keeps the same fundamentals as the benchmark calibrated economy except for two parameters. First, we increase the flow value of the outside option to $\ln(b_U) = 1.79$. A

Table C.1: The effects of the RL in the alternative economy

	No sanctions (Γ=0)	Baseline (Γ=0.04)		No sanctions (Γ=0)	Baseline (Γ=0.04)
A: Orders			*D: Employment*		
Relative price (p_C/p_R)	0.78	0.80	Employment C	0.18	0.15
Market share (o_C/o_R)	0.24	0.21	Employment R	0.71	0.73
Adjustment factor (φ)	0.73	0.72	Outside jobs	0.11	0.12
B: Wages			*E: Hours*		
Mean log hourly wages C	2.05	2.04	Mean effective hours C	4.62	4.66
Mean log hourly wages R	2.21	2.21	Mean effective hours	3.91	3.88
Labor market tightness	1.85	1.89			
C: Job preferences			*F: Welfare change (%)*		
Cutoff C and R jobs, ϵ^*	2.77	2.69	Consumers		-1.24
Cutoff C and U jobs, ϵ^{**}	5.99	5.99	Riders		0.01

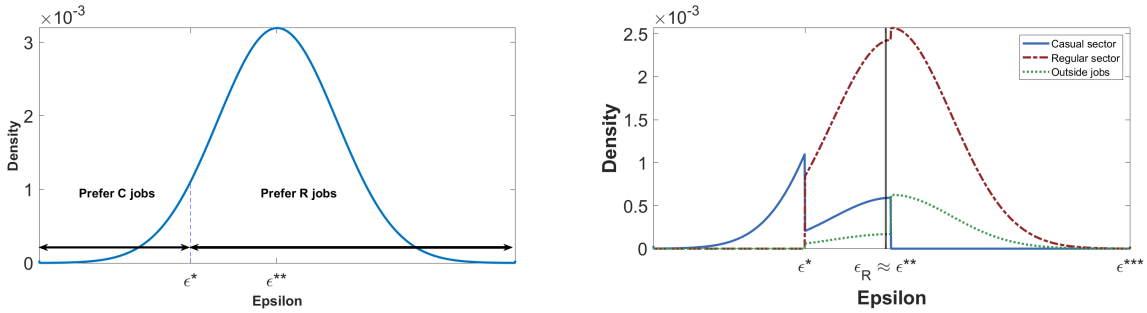
Note: This Table displays the model results from the counterfactual simulation where defier firms in the C sector do not have to pay a fine: $\Gamma = 0$. This is the economy where the parameterization remains the same as in the benchmark except for $\ln(b_U) = 1.79$ and $\iota = 0.45$.

high outside option implies that nearly all riders in the C platform have effective (φh_C) and actual (h_C) hours above those in the R platform before the reform, as riders with a high preference for leisure choose outside jobs rather than working in the C platform. As discussed in Section 4.1, this contributes to a small increase in waiting times when sanctions are in place. Specifically, if riders who switch platforms work more effective hours in R than they did in C before the reform, then they produce more orders. As sanctions do not increase total orders in the food delivery sector, then the remaining C riders must become less productive, i.e., their waiting time increases.[20] Second, to ensure that riders still choose the R platform under a higher outside option, the value of these jobs must increase. This is achieved by reducing search frictions and setting $\iota = 0.45$. We note that this new parameterization does not alter two key features of the benchmark economy where the RL is still detrimental: (i) the benchmark wage premium of 0.17 in the R platform relative to the C platform, and (ii) total employment in the food delivery sector still declines by only 1 percentage point following the reform.

Before discussing the counterfactual results, we first show that the data reject this alternative view of the market. To see this, Figure C.1 displays the distribution of riders' preferences for jobs and the resulting employment distribution in this alternative economy. As mentioned above, increasing the outside option implies that: (i) all C

[20]Only two forces may counteract this channel. First, waiting times do not increase because some other C riders leave the food delivery sector. Second, waiting times do not increase because average hours among the remaining C riders fall. None of these channels are strong enough in our simulations.

Figure C.1: Riders' preferences and employment distribution in the alternative economy



(a) Policy function

(b) Distribution over jobs

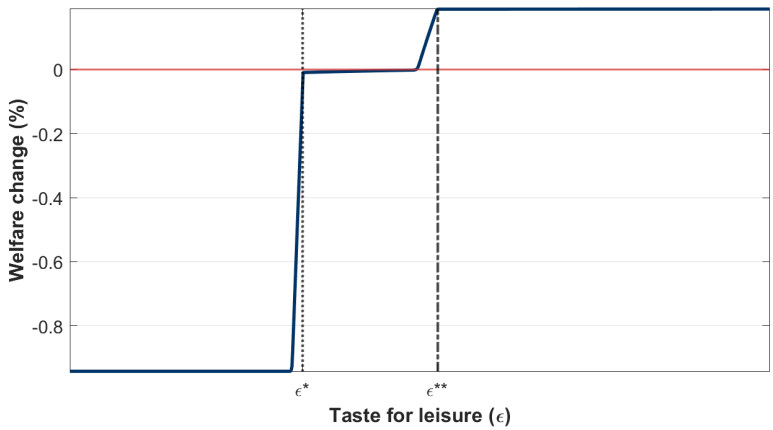
Source: Own elaboration from simulated data.
Note: The left panel displays workers' policies as a function of their time preferences. The right panel displays the employment distributions of the two sectors over workers' time preferences. Note that this is the economy where the parameterization remains the same as in the benchmark except for $\ln(b_U) = 1.79$ and $\iota = 0.45$.

riders work longer hours than their R counterparts, i.e., they have $\epsilon < \epsilon_R$, and (ii) the C platform shrinks. Fact (i) implies that hours worked do not overlap across platforms. Fact (ii) implies that R contracts constitute the dominant form of employment. However, the data show that neither feature holds in the food delivery sector.

Next, Table C.1 reports the labor market effects of the RL in this alternative model economy. The RL induces the reallocation of consumption across platforms by increasing the relative prices of orders served by the C platform. Since the relative decline in hours worked and orders almost match each other, the waiting time hardly changes. In particular, the adjustment factor (φ) and wages in the C platform only decrease by 0.9 percent. Moreover, the wage in the R platform even increases by 0.2 percent because this platform expands, and the outside opportunities of R riders do not deteriorate enough. Overall, the market structure implies that the wage gains in the R sector compensate for the losses in the C sector, so average welfare increases by 0.1 percent. In other words, the RL is solving a clear inefficiency: riders could work similar hours at higher wages. However, for this to hold, the market structure observed in the data should be completely different to what this simulation exercise predicts.

Lastly, we stress that our result on the heterogeneous effects of the RL is robust. Specifically, we define three groups of riders based on the following cutoff values: (i) ϵ^*, such that $W_R^{\text{NoRL}}(\epsilon^*) = W_C^{\text{NoRL}}(\epsilon^*)$, and (ii) $U^{\text{NoRL}}(\epsilon^{**}) = W_C^{\text{NoRL}}(\epsilon^{**})$, where the superscript "NoRL" in the riders' value functions indicates that the economy operates without sanctioning the defiers. Figure C.2 displays the welfare change for the three groups. First, the fraction of riders remaining in the C sector subject to sanctions ($\epsilon < \epsilon^*$) are worse off due to their wage losses. Second, those C riders who search for R jobs ($\epsilon \in [\epsilon^*, \epsilon^{**}]$) remain indifferent between the economies with and without sanctions, as the wage gains in the R platform offset the wage losses in C jobs. Third, those workers

Figure C.2: Welfare effects of the Riders' Law in the alternative economy



Note: The left panel of this Figure displays the welfare effect of the Riders' Law relative to the economy without sanctions against workers' taste for leisure. This is the economy where the parameterization remains the same as in the benchmark except for $\ln(b_U) = 1.79$ and $\iota = 0.45$.

searching for R jobs from outside jobs ($\epsilon > \epsilon^{**}$) are better off due to the wage gains in this sector.